CASE STUDY

Title: Face Insight AI – Smart Age & Gender Detection with AI Chatbot

Team Members

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2. M. Harshitha

Overview

Face Insight AI is a web-based application that combines Computer Vision and Generative AI. The system detects a user's face through a webcam, predicts age and gender, and generates intelligent responses using the Llama 3 chatbot through the Groq API.

Problem Statement

Traditional chatbots provide generic responses and do not understand user characteristics. This project aims to improve user interaction by adding visual awareness and context-based communication.

Solution

The system captures live video, detects faces, predicts age and gender, and sends this information along with the user's query to the AI model. Based on the context, the chatbot generates personalized responses.

Technologies Used

- Python
- FastAPI
- OpenCV
- Groq API
- Llama 3
- HTML, CSS, JavaScript

Team Contribution

M. Pushpavathi

- Participated in project design and implementation
- Assisted in face detection and age/gender prediction testing
- Contributed to frontend interface development
- Prepared project documentation and presentation

M. Harshitha

- Assisted in AI chatbot integration and system development
- Worked on frontend design and user interaction features
- Participated in system testing and debugging
- Contributed to project documentation and presentation

Outcome

The project successfully performs real-time face detection, age and gender prediction, and context-aware AI conversations through a user-friendly interface.

Conclusion

Face Insight AI demonstrates the integration of Computer Vision and Generative AI to build an intelligent and personalized smart assistant.